

NLP Lab: Autogen

Progress Update 21.06.24

What was done [1/3]

Demos are deployed at <http://130.162.48.238:8080/>

Conducted experiments:

- Decision Making: MiniWob++ Experiment
- Coding

Added demos:

- PDF Triaging (simplified version)
- Separate agent for browsing
 - Google Search
 - Webpage retriever via Selenium

What was done [2/3]

Overarching agent

- Based on a groupchat
- Specialised agents with functions as tools
- Fallback to a common text agent
- Agents are not included in the chat unless required (PDF/image)
- Limited functions are included at the moment, more to come
 - PDF/image context
 - Browser access
 - Text responses

What was done [3/3]

Multi-agent Coding experiment finished - needs to deploy on the demo soon though

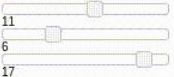
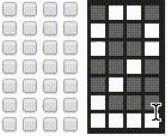
Agent Evaluation (AgentBench), including datasets

- Code (Q&A, Database, Knowledge Graph)
- Game (Card Game, Puzzles, House-Holding)
- Web (Shopping, Browsing)

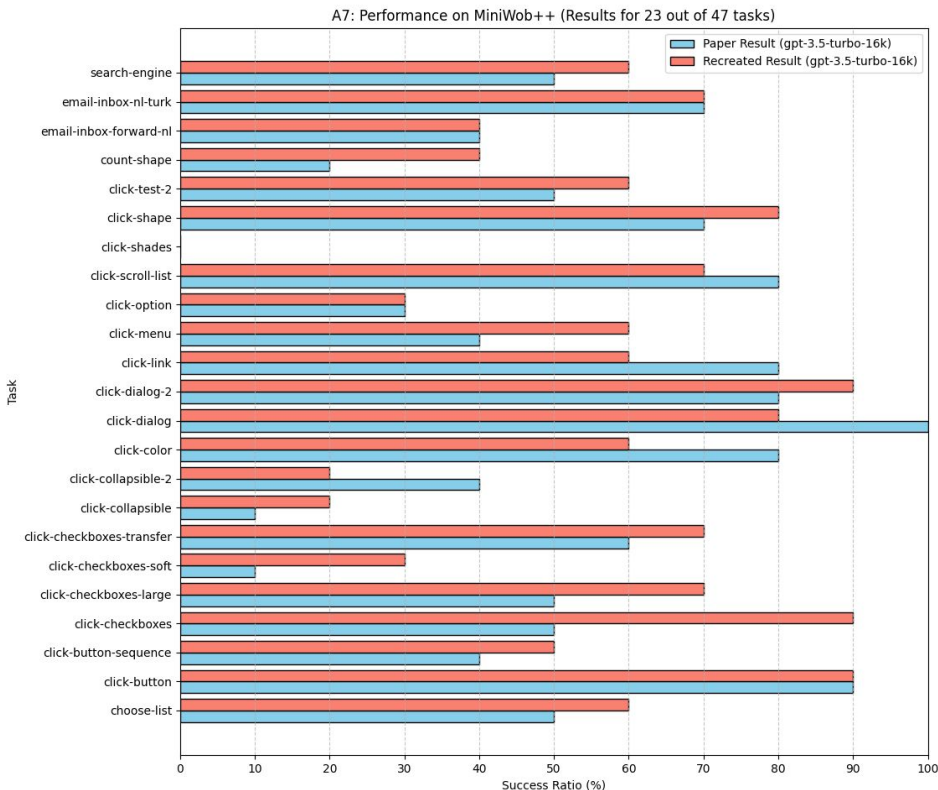
Next steps

- Extending functionality for the overarching agent and find datasets
- Further search for additional experiments
- UI unification
- Bugfixes bugfixes BUGFIXES

Appendix: MiniWeb++ Experiment

Move the cube around so that "5" is the active side facing the user.  Submit	Set the sliders to the combination [13,20,13] and submit.  Submit	Draw the number "2" in the checkboxes using the example on the right and press Submit when finished.  Submit	Select 5 with the spinner and hit Submit. Select a value: 0 Submit	Keep your mouse inside the circle as it moves around. 	Move the cube around so that "4" is the active side facing the user.  Submit
Copy the text in the textarea below and paste it into the textbox. Enim. Elementum, tortor ullamcor Submit	Select 09/23/2016 as the date and hit submit. Date: Submit	Drag all rectangles into the black box.  Submit	Enter "qu" into the text field and press Submit. qu Submit	Select all the shades of blue and press Submit.  Submit	Find the 4th word in the paragraph, type that into the textbox and press "Submit". Non arcu ut ultricies est. Gravida gravida. Porta erat nulla eget condimentum posuere a Submit
Enter an item that starts with "Tuni". Tags: Submit	Enter "Vb8" into the text field and press Submit. Submit	Focus into the 1st input textbox.	Focus into the textbox.	Move the cube around so that "2" is the active side facing the user.  Submit	Select 5Gi and click Submit. 5Gi sPUT Submit

Appendix: MiniWob++ Experiment Results (Our plot)



Average paper success ratio on these 23 out of 47 tasks: 51.74%

Average recreated success ratio on these 23 out of 47 tasks: 56.52%

Appendix: MiniWob++ Remarks

- They use RCI frameworks (<https://arxiv.org/pdf/2303.17491v1>) different base prompts / in-context learning examples for every task for the MiniWob++ agent e.g. for “click-checkboxes large” and “navigate tree” (example.txt):

Example plans)

The task is to click the checkboxes labeled "PtM", "H0t1p0w", "fKWS", "moTD0Z", "LC", "1QDslel", "LCVeO5", "PqN" and then click the "Submit" button.
Here is a plan to solve this example task on the webpage with the autonomous agent.

1. clickxpath //label[text() = 'PtM']/input
2. clickxpath //label[text() = 'H0t1p0w']/input
3. clickxpath //label[text() = 'fKWS']/input
4. clickxpath //label[text() = 'moTD0Z']/input
5. clickxpath //label[text() = 'LC']/input
6. clickxpath //label[text() = '1QDslel']/input
7. clickxpath //label[text() = 'LCVeO5']/input
8. clickxpath //label[text() = 'PqN']/input
9. clickxpath //button[@id='subbtn']

Example plans)

task: navigate through the file tree. Find and click on the folder or file named "Riley".
plan:

1. clickxpath //span[text()='Briana'] // Click the expandable hitarea of the folder "Briana"
2. clickxpath //span[text()='Riley']

Appendix: MiniWob++ Experiment Results (Paper plot)

